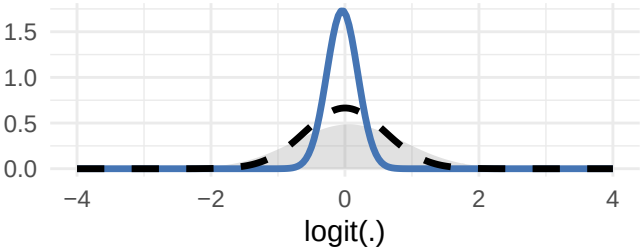


Population-Level Recovery and Variance Shrinkage

Population Distribution: logit(.)

Notice how the unpooled estimates are artificially too wide.

Density

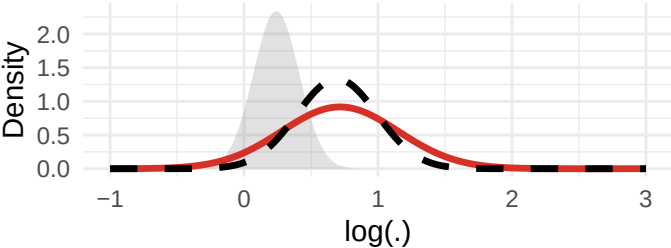


— Hierarchical — True Unpooled Estimates

Population Distribution: log(.)

The hierarchical model recovers the true, tighter distribution.

Density

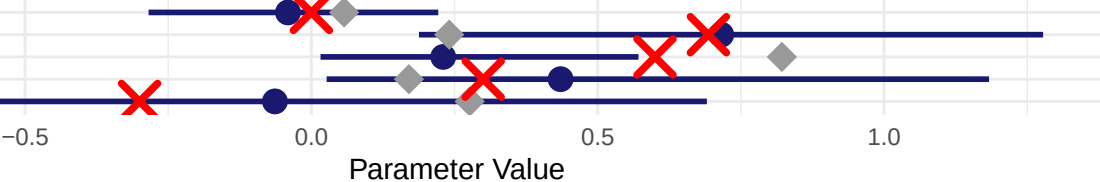


— Hierarchical — True Unpooled Estimates

Hyperparameter Recovery & Partial Pooling

Comparing Hierarchical Posteriors (Mean $\pm$ 90% CI), Unpooled Sample Stats, and True Values

Mean: logit(.)
Mean: log(.)
SD: logit(.)
SD: log(.)
Correlation: .



● Hierarchical X True ◆ Unpooled